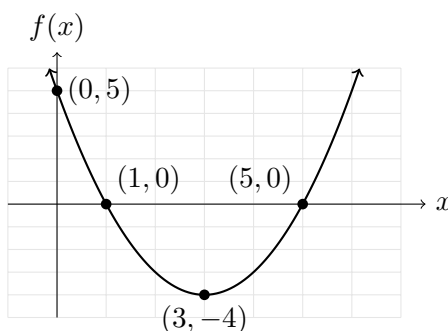


Midterm 1 Review: Solution Key

Answers follow the current review sheet. In Problem 7(a), “average value” means the average of the two endpoint values.

1. Quadratics and constraints.

- (a) $f(x) = (x - 3)^2 - 4 = (x - 1)(x - 5)$. The vertex is $(3, -4)$; the x -intercepts are $(1, 0)$ and $(5, 0)$, and the y -intercept is $(0, 5)$. The parabola opens upward, with axis $x = 3$.



- (b) On $[0, 5]$, compare $f(0) = 5$, $f(3) = -4$, and $f(5) = 0$. The maximum is 5 at $x = 0$; the minimum is -4 at $x = 3$.
- (c) On $[4, 7]$, the function is increasing. The minimum is $f(4) = -3$ and the maximum is $f(7) = 12$. The vertex is outside this interval.

2. Reading a graph.

- (a) In the order asked: -2 , 4 , 3 , -1 , -2 , and undefined.
- (b) Domain: $[-4, 6)$. Range: $[-2, 4]$.
- (c) Absolute maximum: 4 at $x = -1$. Absolute minimum: -2 at both $x = -4$ and $x = 4$.
- (d) Increasing on $(-4, -1)$ and $(2, 4)$; decreasing on $(-1, 2)$ and $(4, 6)$.
- (e) In the order asked: 4 , 1 , -1 , DNE, 1 . The limit at 2 fails because its one-sided limits differ.
- (f) $g(2) = -1$ equals the right-hand limit, but the left-hand limit is 1 . A function value describes the filled point; a limit describes nearby values.
- (g) Both one-sided limits at 4 and the two-sided limit equal 3 , whereas $g(4) = -2$. This is a removable discontinuity.

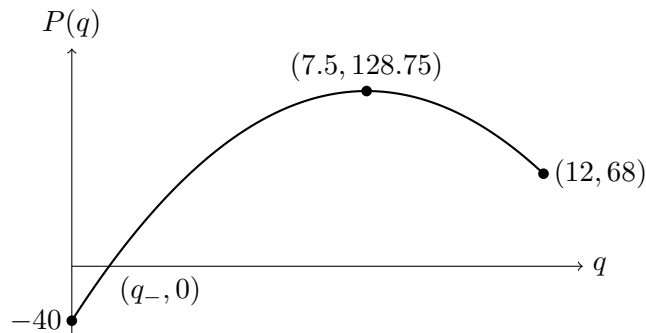
3. **Domains.** (a) $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$, since $x^2 - 9 \neq 0$. (b) $(-\infty, 7/2)$, since $7 - 2x > 0$. (c) $[-2, \infty)$, since $3x + 6 \geq 0$. (d) $(4, \infty)$, since the denominator requires $x - 4 > 0$.

4. Cost, revenue, profit, and whole-number quantities.

- (a) $C(q) = 40 + 7q$, $R(q) = q(52 - 3q) = 52q - 3q^2$, and $P(q) = R(q) - C(q) = -3q^2 + 45q - 40$, for $0 \leq q \leq 12$.
- (b) Completing the square gives $P(q) = -3(q - 15/2)^2 + 515/4$. The vertex is $(15/2, 515/4) = (7.5, 128.75)$ and the vertical intercept is $(0, -40)$. The algebraic roots are

$$q_- = \frac{45 - \sqrt{1545}}{6} \approx 0.9489, \quad q_+ = \frac{45 + \sqrt{1545}}{6} \approx 14.0511.$$

Only $(q_-, 0)$ is on the graph restricted to $[0, 12]$.



- (c) Compare $P(0) = -40$, $P(7.5) = 128.75$, and $P(12) = 68$. Maximum: \$128.75 at $q = 7.5$ in the real model. Minimum: -\$40 at $q = 0$.
- (d) The algebraic break-even quantities are q_- and q_+ above, where revenue equals cost. The capacity constraint excludes q_+ , so only q_- is feasible. Algebraically, profit is positive for $q_- < q < q_+$; on the stated domain this becomes $q \in (q_-, 12]$.
- (e) Compare the integers adjacent to 7.5: $P(7) = P(8) = 128$. Thus both 7 and 8 baskets maximize profit, at \$128. Every whole-number quantity $1, 2, \dots, 12$ earns positive profit.

5. Supply and demand.

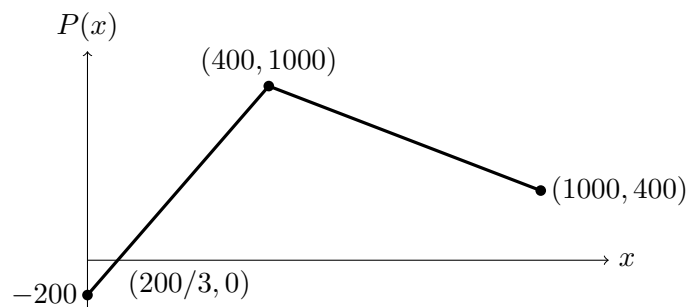
- (a) The demand slope is $(16 - 10)/(60 - 120) = -0.1$. Using $(q, p) = (120, 10)$ gives $p_D(q) = 22 - 0.1q$.
- (b) The domain is $[0, 220]$. Each additional unit of weekly demand corresponds to a \$0.10 decrease in price per unit. The vertical intercept is \$22 per unit, the modeled price at zero demand.
- (c) $22 - 0.1q = 4 + 0.05q$ gives $q = 120$ units per week and $p = \$10$ per unit.
- (d) $R(q) = 22q - 0.1q^2$ on $[0, 220]$. Its vertex occurs at $q = -22/[2(-0.1)] = 110$. The corresponding price is $p_D(110) = \$11$ and maximum revenue is $R(110) = \$1,210$ per week.

6. Building and maximizing a piecewise model.

- (a) The first branch is $3x - 200$, so $P(400) = 1000$. The second branch decreases from this value:

$$P(x) = \begin{cases} 3x - 200, & 0 \leq x \leq 400, \\ 1400 - x, & 400 < x \leq 1000. \end{cases} \quad \text{Domain: } [0, 1000].$$

- (b) Connect $(0, -200)$ to $(400, 1000)$ to $(1000, 400)$. The intercepts are $(0, -200)$ and $(200/3, 0)$. All endpoints are included.



- (c) The maximum profit is \$1,000 at an advertising budget of \$400.
- (d) $P(200) = 400$, $P(800) = 600$, so the average rate is $(600 - 400)/(800 - 200) = 1/3$ dollar of profit per dollar of advertising. Over this interval, profit increases by about \$0.33 per additional advertising dollar on average.
- (e) $y - 400 = \frac{1}{3}(x - 200)$, or $y = \frac{1}{3}x + \frac{1000}{3}$.
- (f) All three limits at 400 equal 1000.

7. Average rates of change and secant lines.

- (a) $f(1) = 0$, $f(4) = 6$. The average of the endpoint values is $(0 + 6)/2 = 3$.
- (b) The average rate of change is $(6 - 0)/(4 - 1) = 2$. Part (a) averages two output values; part (b) measures the change in output per unit change in input.
- (c) $y - 0 = 2(x - 1)$, or $y = 2x - 2$.
- (d) $f(x + h) - f(x) = 2xh + h^2 - 3h$, so the quotient is $2x + h - 3$, $h \neq 0$.

8. Marginal cost, revenue, and profit.

- (a) $P(q) = -q^2 + 24q - 100$. Thus $P(5) = -\$5$ is a \$5 loss when 5 units are sold, and $P(6) = \$8$ is an \$8 profit when 6 units are sold.
- (b) $C(5 + h) - C(5) = 16h + h^2$, so $M_C(h) = 16 + h$. Thus $M_C(5) = 21$, $M_C(3) = 19$, and $M_C(1) = 17$, in dollars per unit. These are the average cost increases per additional unit from 5 to 10, 5 to 8, and 5 to 6 units, respectively.
- (c) The marginal cost is $C(5) - C(4) = \$15$ is the cost of producing the 5-th unit. The marginal revenue is $R(5) - R(4) = \$30$ is the revenue from selling the 5-th unit. The marginal profit is $P(5) - P(4) = \$15$ is the profit from selling the 5-th unit.

9. Exponential models and thresholds.

- (a) $V(t) = 12000(0.85)^t$, $t \geq 0$. $V(2) = 12000(0.85)^2 = \$8,670$, the machine's value two years after purchase.
- (b) $[V(2) - V(0)]/2 = (8670 - 12000)/2 = -1665$ dollars per year. Over the first two years, the value decreases by \$1,665 per year on average.
- (c) $12000(0.85)^t = 6000$ gives

$$t = \frac{\ln(1/2)}{\ln(0.85)} \approx 4.27 \text{ years.}$$

The value is below \$6,000 precisely when $t > \ln(1/2)/\ln(0.85)$. The inequality reverses when dividing by $\ln(0.85) < 0$.

- (d) $N(t) = 800(1.12)^t$. Then

$$N(t) > 1500 \iff t > \frac{\ln(15/8)}{\ln(1.12)} \approx 5.55 \text{ years.}$$

The first whole-year anniversary with more than 1,500 subscribers is year 6.

10. Exponential and logarithmic equations.

- (a) $(2x - 1) \ln 3 = \ln 20$, so $x = \frac{1}{2} \left(1 + \frac{\ln 20}{\ln 3}\right)$.
- (b) $e^{-t/5} = 1/4$, so $t = -5 \ln(1/4) = 5 \ln 4$.
- (c) Both logarithms require $x > 1$. Combining them gives $\ln(x^2 - 1) = \ln 8$, hence $x^2 = 9$. Only $x = 3$ is allowed; $x = -3$ violates the original domain.
- (d) $x - 3 = 2^4 = 16$, so $x = 19$, which satisfies $x > 3$.

11. Basic limits and limit laws. All expressions in this problem are interpreted as limits as $x \rightarrow a$, including those with an omitted limit symbol on the review sheet.

- (a) In order, the limits are

$$2, \quad 12, \quad -8, \quad -2, \quad 16, \quad -\sqrt[3]{2}.$$

Use the sum, constant-multiple, product, quotient, power, and cube-root laws, respectively. The quotient law applies because $g(x) \rightarrow -2 \neq 0$.

- (b) The four answers, in order, are: *not enough information*, -6 (finite), 2 (finite), and *not enough information*. For the middle two expressions, direct application of the limit laws gives

$$\frac{4^2 - 4}{-2} = -6, \quad \frac{(-2)^2 + 4}{4} = 2.$$

For the first expression, where defined,

$$\frac{g(x) - 2}{4 - g(x)^2} = -\frac{1}{g(x) + 2}.$$

Its numerator does not approach zero before cancellation, and its denominator approaches zero. The sign of approach is unspecified: choosing $g(x) = -2 + (x - a)^2$ gives $-\infty$; choosing $g(x) = -2 - (x - a)^2$ gives $+\infty$; choosing $g(x) = -2 + (x - a)$ gives opposite one-sided infinities.

Likewise, $g(x)^2/[f(x) - 4]$ has numerator tending to 4 and denominator tending to 0. With $g(x) = -2$, the choices $f(x) = 4 + (x - a)^2$, $f(x) = 4 - (x - a)^2$, and $f(x) = 4 + (x - a)$ give $+\infty$, $-\infty$, and a nonexistent two-sided limit, respectively. Neither expression can have a finite limit along its domain as the denominator approaches zero.

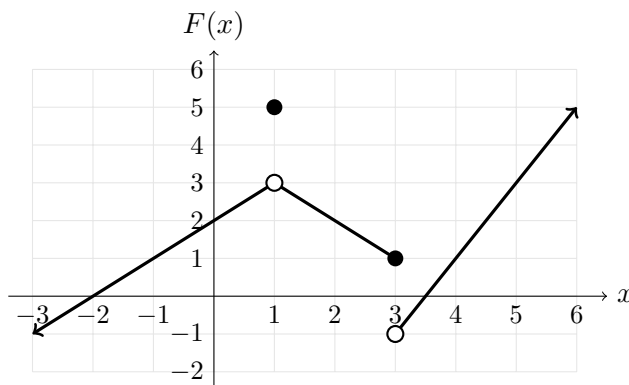
12. **Evaluate the following limits.** By continuity, substitute the limiting input: (a) $3(2)^2 - 2 + 4 = 14$; (b) $\sqrt{5+4} = 3$; (c) $2^0 = 1$; (d) $\ln e = 1$.

13. **Rational-function limits.**

- (a) Direct substitution gives $(2^2 + 3)/(2 + 1) = 7/3$.
 (b) For $x \neq 3$, $(x^2 - 9)/(x - 3) = x + 3$. The limit is 6, although the original function is undefined at 3.
 (c) For $x \neq -2$, $(x^2 + x - 2)/(x + 2) = x - 1$. The limit is -3 , although the original function is undefined at -2 .
 (d) The left-hand limit is $-\infty$ and the right-hand limit is $+\infty$. The two-sided limit does not exist, and the function is undefined at 1.

14. **Piecewise functions and limits.**

- (a) Draw $y = x + 2$ for $x < 1$, with an open circle at $(1, 3)$; a filled point at $(1, 5)$; $y = 4 - x$ for $1 < x \leq 3$, open at $(1, 3)$ and closed at $(3, 1)$; and $y = 2x - 7$ for $x > 3$, open at $(3, -1)$. Thus $F(1) = 5$ and $F(3) = 1$.



- (b) At $x = 1$: $\lim_{x \rightarrow 1^-} F(x) = 3$, $\lim_{x \rightarrow 1^+} F(x) = 3$, and $\lim_{x \rightarrow 1} F(x) = 3$. At $x = 3$: $\lim_{x \rightarrow 3^-} F(x) = 1$ and $\lim_{x \rightarrow 3^+} F(x) = -1$, so $\lim_{x \rightarrow 3} F(x)$ does not exist.
 (c) Set $F(1) = 3$. Changing only $F(3)$ cannot make the limit at 3 exist: the one-sided limits remain 1 and -1 regardless of the value assigned at the point itself.

15. Reading a more complicated graph.

- (a) Increasing on $(-4, -3)$, $(-1, 1)$, and $(1, 3)$. Decreasing on $(-3, -1)$ and $(3, 5)$. The intervals cannot cross the vertical asymptote at $x = 1$.
- (b) Concave up on $(-2, 1)$; concave down on $(-4, -2)$ and $(1, 5)$. The only inflection point is $(-2, 2)$. Although the concavity differs across $x = 1$, that input is outside the domain and is not an inflection point.
- (c) Compare all turning points and endpoints within each interval:

Interval	Absolute maximum		Absolute minimum	
	Value	All locations	Value	All locations
$[-4, -1]$	4	$x = -3$	0	$x = -4, -1$
$[-2, 0]$	4	$x = 0$	0	$x = -1$
$[-1, 0]$	4	$x = 0$	0	$x = -1$
$[2, 5]$	4	$x = 3$	2	$x = 2, 5$
$[2, 3]$	4	$x = 3$	2	$x = 2$

- (d) H is not defined on the whole interval $[-1, 2]$, since $H(1)$ is undefined. Even on the allowed inputs $[-1, 1) \cup (1, 2]$, there is neither an absolute maximum nor an absolute minimum: the function is unbounded above as $x \rightarrow 1^-$ and unbounded below as $x \rightarrow 1^+$.
- (e) In the order asked,

$$+\infty, \quad -\infty, \quad \text{DNE}, \quad 2, \quad 4.$$

The two-sided limit at 1 does not exist because the one-sided limits have opposite infinite behavior. The limits at -2 and 3 agree with the graph's values there.